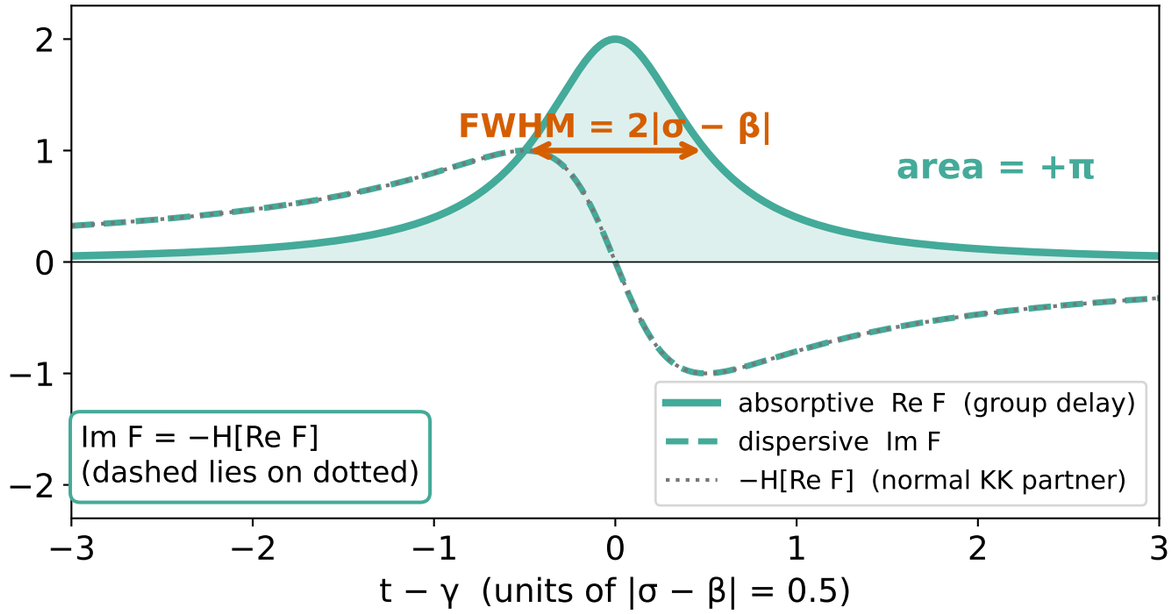
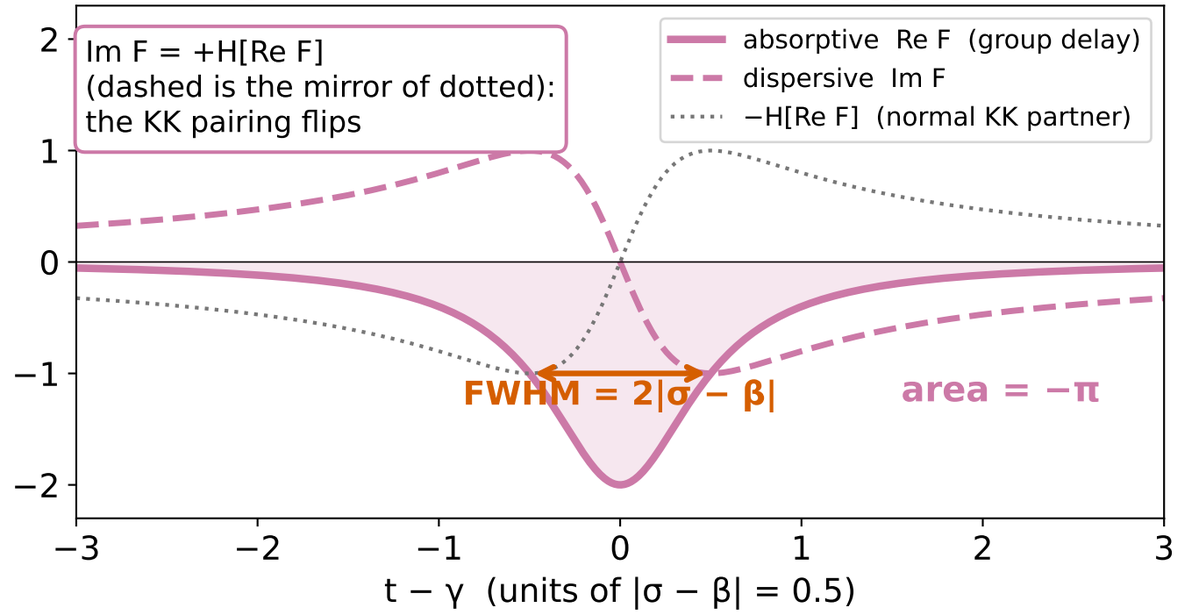
The Kramers–Kronig pair of a zero: absorption vs gain, and Γ as the full width at half maximum

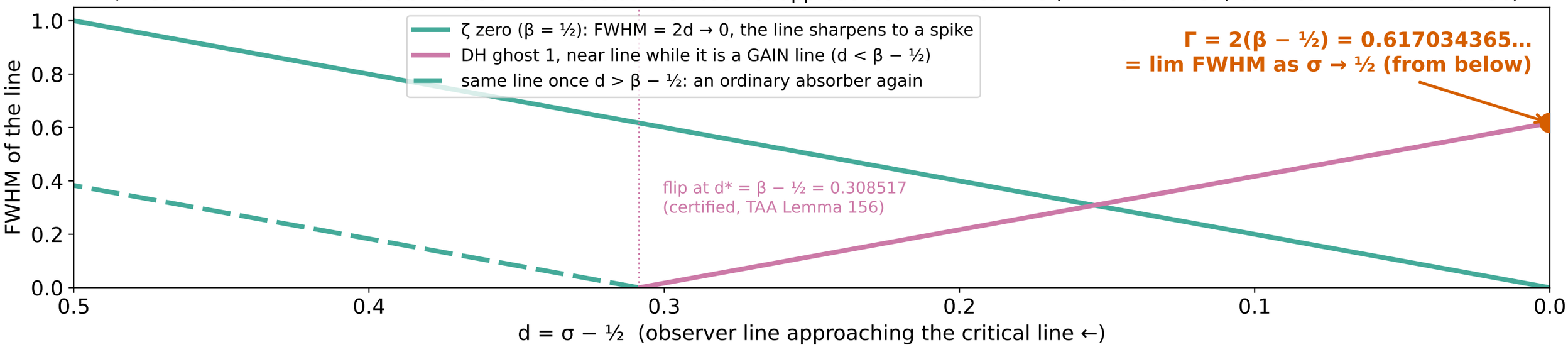
Absorption line (ζ-type: the observer line σ is right of the zero, σ > β)



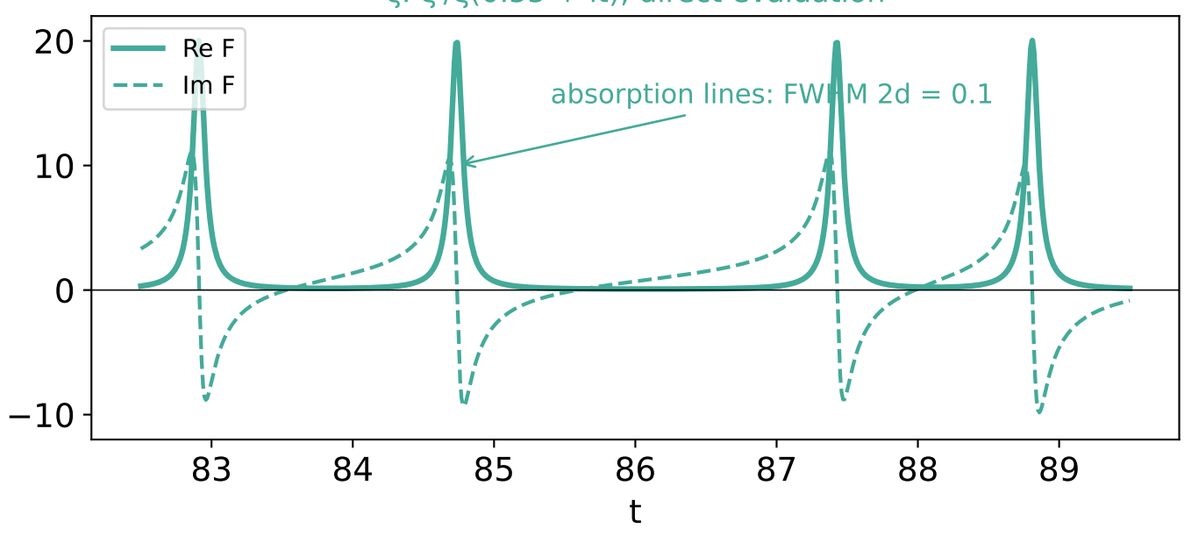
Gain line (DH ghost seen from σ < β: the zero is right of the observer)



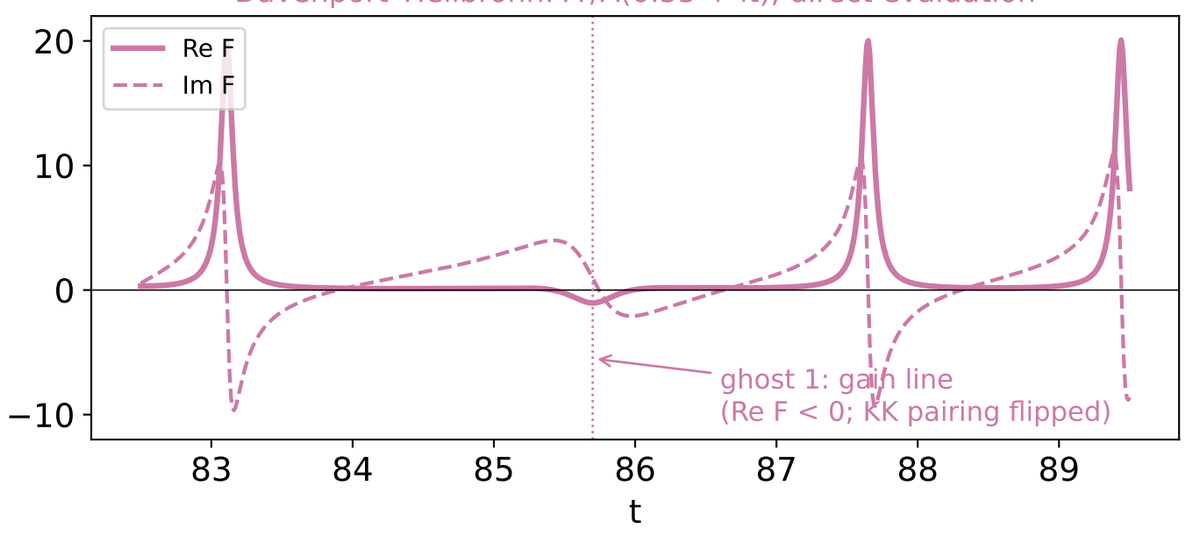
Γ, the width of a zero: what each line's FWHM does as the observer approaches the critical line (TAA Lemma 154, certified in TAA Lemma 156)



ζ: ξ′/ξ(0.55 + it), direct evaluation



Davenport–Heilbronn: Λ′/Λ(0.55 + it), direct evaluation



Theorem (Hadamard; TAA Lemma 154): on Re s = σ each zero ρ = β + iγ contributes (σ−β)/((σ−β)² + (t−γ)²) to Re F and −(t−γ)/((σ−β)² + (t−γ)²) to Im F, so Im F = −sgn(σ−β)·H[Re F] line by line. RH ⇔ no gain line on any σ > ½ (Lagarias). DH data: direct evaluations from TAA Lemma 154; β from the TAA Lemma 156 Arb certificate. RH is not proved.

Colours from the atlas concept palette: absorption line = 'resonance' #44AA99 (teal (Tol muted)),
gain line / ghost = 'ghost' #CC79A7 (reddish purple (Okabe–Ito)), Γ / critical-line distance = 'critical_line' #D55E00 (vermillion (Okabe–Ito)); grey = Hilbert transform reference.